EXPERIMENT 6

IMPLEMENTATION OF TREE TOPOLOGY USING PACKET TRACER

Aim

To implement a tree topology using Packet Tracer and to transmit data between the devices connected in the tree topology.

Software Used

Cisco Packet Tracer (end devices, switches, connecting cables)

Procedure

Step 1: Start Cisco Packet Tracer.

Step 2: Click **End Devices**, select the Generic PC and place four PCs in the workspace.

Step 3: Click **Switches** and place three switches in the workspace: one root switch (Switch0) and two lower-level switches (Switch1 and Switch2).

Step 4: Click **Connections**, select the Copper Straight-Through cable and connect two PCs to Switch1 and two PCs to Switch2. This creates two star topologies.

Step 5: Connect Switch1 and Switch2 to the root switch Switch0. This joins the star topologies into a tree topology.

Step 6: Click each PC, open the **Config** tab, select FastEthernet0 and type the IP address. The subnet mask is generated automatically.

Step 7: In **Realtime** mode, select the **Add Simple PDU** tool and click PC0 and then PC3. The PDU Last Status should show **Successful**.

Step 8: In **Simulation** mode, keep only the ICMP filter, send the PDU from PC0 to PC3 and click Capture/Forward until the ping is completed. The ICMP packet moves through Switch1, Switch0 and Switch2 and the status shows **Successful**.

Execution Screenshot

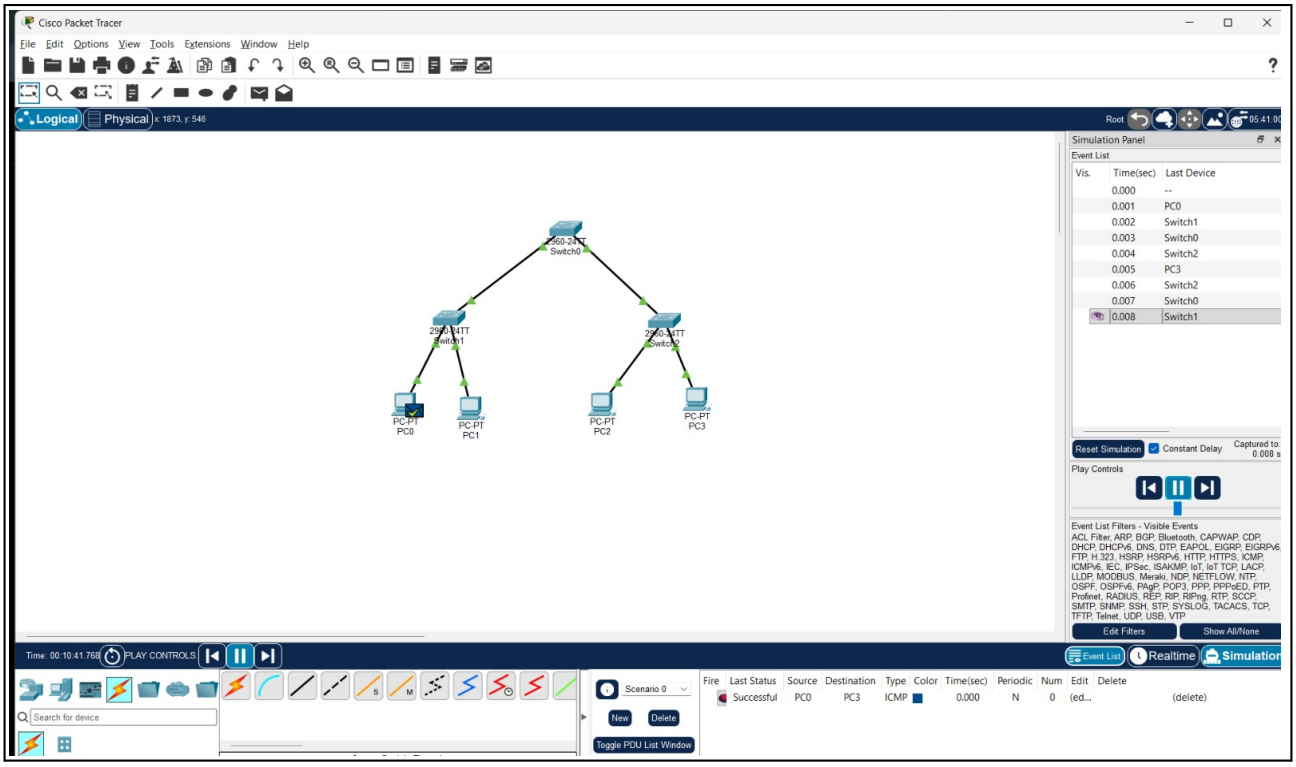


Fig: Tree topology with ICMP ping from PC0 to PC3

Result

Thus the tree topology is implemented with the Packet Tracer simulation tool.